



DESCRIPTION OF FINAL ASSESSMENT

Course Code	AIT202
Course Name	Methods and Applications of Deep Learning
Lecturer	Raheel Zafar
Academic Session	202504
Assessment Title	Deep Learning for Practical Problems: A Group Project

A. Introduction

Machine learning classification is the task of assigning an input one label from a fixed set of categories. Machine learning models use algorithms to find the pattern in the data instead of being explicitly programmed with the set of instructions. Deep learning which is a branch of machine learning, allows machines to identify and extract features from images, text, signals, which helps to classify accurately. The convolutional neural network (CNN) is a well-recognized deep learning architecture used for solving various classification problems. The architecture of a CNN is analogous to that of the connectivity pattern of neurons in the human brain and was inspired by the organization of the visual cortex.

Objectives

- Understand the working of CNNs for image classification.
- Train a model to recognize facial expressions and classify them into emotions like happy, sad, angry, etc.
- Evaluate model performance using metrics like accuracy, precision, recall, and F1-score.
- Visualize results and misclassifications using confusion matrices and Grad-CAM.

Deep Learning Concepts Used

- Convolutional Neural Networks (CNNs)
- Activation functions (ReLU, Softmax)
- Loss functions (Cross-Entropy)
- Optimization algorithms (Adam, SGD)

- Data Augmentation
- Transfer Learning (optional for higher accuracy using pre-trained models like VGG16 or ResNet)

Software Requirements:

- **Language:** Python
- **Libraries:** TensorFlow / Keras or PyTorch, NumPy, Matplotlib, OpenCV
- **Tools:** Jupyter Notebook / Google Colab

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 3 Construct deep learning methods for practical problems within a development environment such as Tensorflow and PyTorch. (P5, PLO3).

CLO 4 Complete a project involving deep learning methods with team work and communication skills. (A4, PLO5).

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
 - i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
 - ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
 - iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.
2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

You are required to organize a group of **5 - 6 members** to do a group task, which requires you to work together to **create a dataset, implement a complex deep learning model using PyTorch, and write a report**. You should **divide the work fairly** between the group members. Document the work properly and use figures, especially when commenting on the methodology. The task also requires you to **prepare a presentation of 15-20 minutes** (maximum) where **each individual is given a maximum of four minutes** to illustrate their contribution to the work. At the end of the presentation, you have to give the demonstration of the project.

Prepare your report in **pdf** format and **name the file under your student IDs** (e.g., “AIT2xxx, AIT2xxx, AIT2xxx”). **Submit the pdf file on Turnitin system by Week 12, 11 PM (Sunday, 22nd June)**. Only **one copy should be submitted** (by the group leader). As for the dataset and codes, please upload them to cloud storage and include the links in the report. Make sure the lecturer has permission to download the files from the included links in the pdf. Please test your links before making a submission. **[Turnitin Class ID 48266292, Enrolment key 202504]**. **Each group should take some pictures for their own project (only to Test the Model). Sharing of those pictures is not allowed.**

Late submissions will be accepted within a week after the deadline; however, the final mark will be **reduced** depending on the reason. You may refer to the marking rubrics for the detailed requirement for each part. Please **use the cover page** and **attach the rubrics appendix** together with the submission of the project. Write your details on the cover page.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Coding and Report	70%
2.	Presentation	30%
	TOTAL	100%

F. Tasks

1. Introduction to the Facial Expression Recognition

- Explain the basics of Facial Expression Recognition (what it is and how it works).
- Briefly introduce the dataset, its importance, and the types of objects it contains.

2. Dataset Exploration

- FER2013 (**Facial Expression Recognition**), Download the data set from the following link.
- You may take some Facial expression pictures by yourself and use those pictures for the purpose of testing to see the results of your model.
- <https://www.kaggle.com/datasets/msambare/fer2013>
- **If you have any difficulty in training of data set, please train your model on limited (less) number of features BUT you have to write all the details in the report.**

3. Data Preprocessing

- Resize/normalize images
- Convert grayscale to RGB
- Label encoding
- Split dataset
- Data Augmentation

4. Model Building

- Build a **CNN architecture**:
 - Input layer
 - Convolution + Pooling layers
 - Flatten
 - Dense (Fully Connected) layers
 - Output layer (Softmax for multi-class classification)

Optional: Use Transfer Learning with models like VGG16, ResNet, or MobileNet.

5. Model Training

- Compile model with:
 - Optimizer: Adam
 - Loss function: `categorical_crossentropy`
 - Metric: `accuracy`
- Train using `model.fit()` with train and validation sets.
- Use callbacks like `EarlyStopping` or `ModelCheckpoint`.

6. Model Evaluation

- Evaluate using test set.
- Generate:

- Accuracy score
- Confusion Matrix
- Classification Report (Precision, Recall, F1)
- Loss/Accuracy Curves for training and validation

7. Results and Analysis

- Compare model performance with baseline models or literature.
- Identify misclassified images and analyze why errors occurred.
- Discuss overfitting/underfitting based on training vs. validation accuracy.

8. Analysis guidelines

- Project planning and dataset exploration
- Preprocessing and data augmentation
- Implement basic CNN model
- Train and evaluate the model
- Improve accuracy (tune hyperparameters / use transfer learning)
- Visualize and analyze results
- Write report and prepare presentation

9. Evaluation Metrics

- Accuracy
- Precision, Recall, F1-Score
- Confusion Matrix
- ROC Curve (if applicable)
- Grad-CAM Visualization (to show model interpretability)

10. Expected Deliverables

- Python code / Jupyter notebook
- Trained model
- Project report (PDF or Word): Introduction, Methodology, Results, Conclusion
- Final presentation (PowerPoint)

Literature review is very important in any analysis. You must read and mention the related architectures/methods that you use as a reference to design the model. You will need to justify your selection of architecture. Other related and recent studies may also be reviewed in your literature. A comprehensive and critically written literature may increase your final mark. However, once your model is designed, you

will need to train and test it using the dataset you created. You should explain how you are going to evaluate your model. Report and discuss the initial results of the model (e.g., accuracy). After that, attempt to improve the results where you will be experimenting and documenting the things you applied (including those that made the results get worse).

The written report should include the following (note that the number of words is just a guideline; meaning that it is not a limitation):

- Introduction (100-300 words).
- Literature review (300-2000 words). Citations are required here.
- Methodology and dataset (300-1000 words).
- Results (300-1000 words).
- Conclusion, limitations, and future work (100-300 words).
- Member's contributions (list the contributions for each student).
- References (APA format).
- Dataset and code links (here you put the links for the dataset and code that you uploaded on the cloud).
- Important codes and appendices (only insert the main codes as an appendix. All the other codes should be found in the cloud).
- Note: if your code is not too large, then you can put it all as appendix.

Following steps should be properly mentioned in the methodology section of the report.

- Data Collection and Preprocessing
 - Data Selection
 - Data Preprocessing
- Image feature extraction (with or without Deep learning architecture)
 - Pretrained CNN Model (Explain complete architecture)
 - Feature Extraction
- Sequence Generation (with Deep learning architecture)
 - Complete model architecture
- Model Training
 - Loss function, optimization, training loop, cross validation, split ratio
- Model Evaluation
 - Evaluation metrics [Find all which we have studied]
- Fine tuning and Hyperparameter tuning
- Inference and deployment (if any)

APPENDIX 1: MARKING RUBRICS

Component Title	Coding and Report			Percentage (%)	70%	
Criteria	Score and Descriptors				Weight (%)	Marks
	Excellent (8-10)	Good-Average (4-8)	Need Improvement (0-4)			
Introduction	Clear and precise introduction of the topic, methods, and scope. The introduction gives an impressively clear idea of the work done in the project.	The students introduce the context, and it is fairly clear.	The introduction is not clear as it does not provide enough information to understand what the work of the project is about.		10	
Literature review	The literature review is comprehensive and written critically. It demonstrates an understanding of the related studies and methods.	The literature review is appropriate, and it clearly reviews a few related studies and methods.	The literature review is unorganized with limited information.		10	
Methodology /model and dataset	The dataset collection and preparation process are clearly explained with some dataset analysis. The method is well described with justifications. Some innovative ideas are put into it. In addition, the selection of evaluation measurements is good.	The dataset and method are described appropriately.	The description of the used dataset and method is not clear. It is difficult to understand the method or the dataset preparation process.		10	
Results	The results are illustrated and described clearly, and the figures and tables are appropriately formatted. Some comparisons are made and described.	Training and testing results are provided and somewhat described well.	Some results are provided, but it is not easy to interpret them.		10	
Conclusion, limitations, and future work	A clear and critical conclusion of the project is found. The students are able to point out the findings, limitations, and future work.	The conclusion is somewhat written well. The limitations and future work are given.	The conclusion needs more effort. The limitations or future work are lacking or not included.		10	
Coding	The coding is precise and excellent. All the ideas described in the methodology section are implemented correctly.	The code works and it has a good style.	The code is unorganized and contains unnecessary parts. Also, some codes are not working properly.		10	
Report quality	The students show excellence in writing a clear and precise report. The lecturer's instructions are followed precisely. A comprehensive list of references is found, which are cited in-text and formatted correctly.	The report is reasonably organized and understandable. References are sufficient, but with minor inconsistencies.	The report is unclear/unorganized. The references are not formatted correctly.		10	
TOTAL						
Component Title	Presentation			Percentage (%)	30%	
Criteria	Score and Descriptors				Weight (%)	Marks
	Excellent (8-10)	Good-Average (4-8)	Need Improvement (0-4)			
Presentation quality	The presenters show remarkable skill in demonstrating. The quality is excellent, and the students are able to cover a proper amount of content within the given time limit.	The presenters' sound is clear and understandable. The duration is less than fifteen minutes (in the case of three members).	The quality of the presentation needs improvement. No proper presentation slides are used.		10	
The contents flow and teamwork	The flow of the contents is very easy to follow and well-thought-out. From the presentation, you can see that there is a lot of teamwork effort has been put into preparing the presentation. Also, the students can state/demonstrate their contribution to the project.	The flow of the contents is good and can be followed. The contents/time of the presentation is somehow divided appropriately among the group members.	The flow of the contents needs improvement. Teamwork is lacking.		10	
Presentation contents	The students manage to present all the most important contents of their work such as the dataset, model, results, and findings, which can precisely demonstrate the hard work done to complete the project.	The contents are appropriate, and the students can demonstrate their understanding of these contents.	The contents of the presentation need improvement, and the students show a lack in demonstrating their understanding of what is being presented. Some technical terms and descriptions are incorrect.		10	
TOTAL						

Note to students: Please print out and attach this appendix together with the submission of coursework